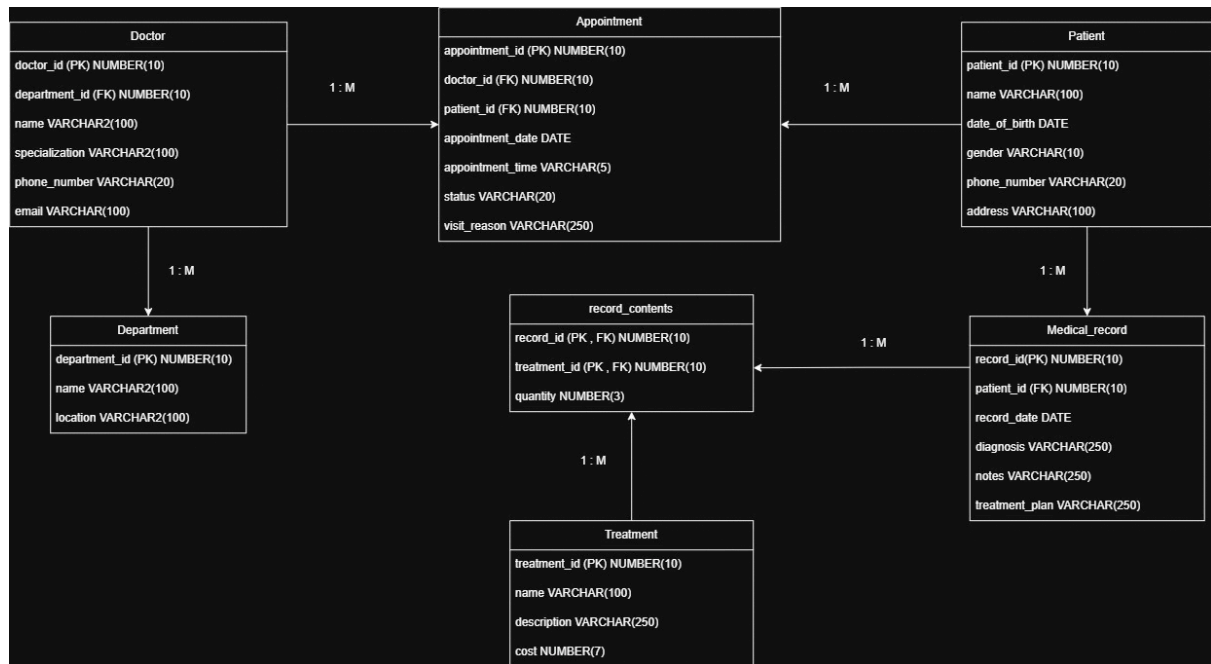


Hospital Management System



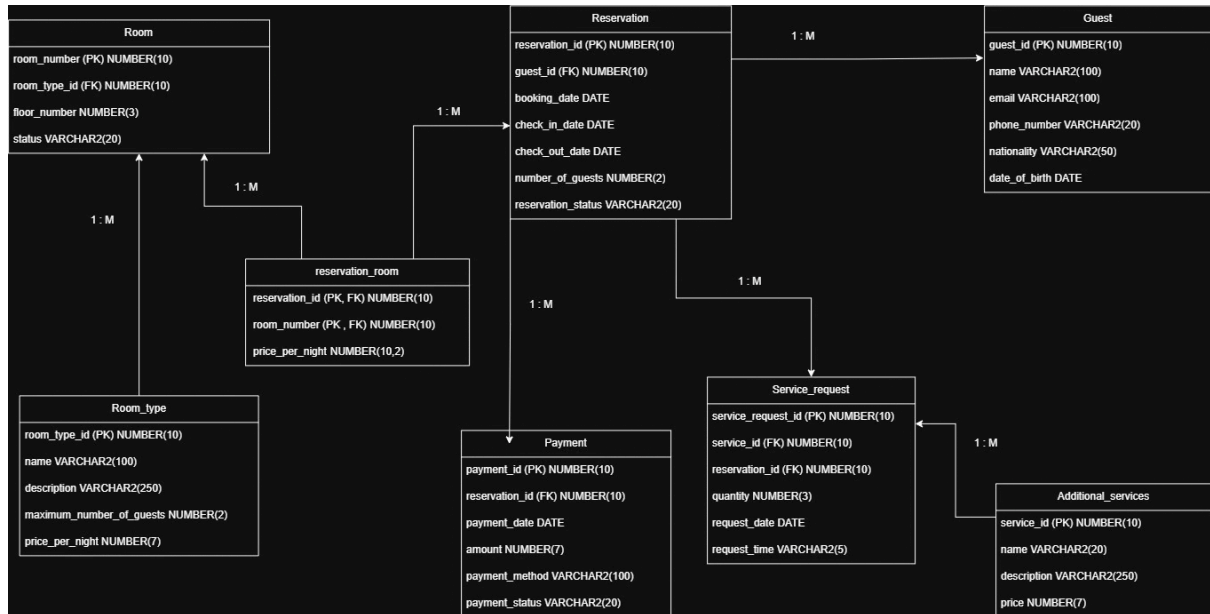
1. How does your ERD represent the relationship between patients and doctors through appointments?

Appointments is a junction table to resolve the M:N relation between the tables Patient and Doctor, it has doctor_id and patient_id as foreign keys and it has its own primary key (appointment_id) because a patient can see the same doctor many times, so the pair doctor_id and patient_id can't identify a row.

2. How does your ERD represent the relationship between medical records and treatments?

Record_contents is a junction table to resolve the M:N relation between the tables Medical_record and Treatment, it has a composite primary key consisting of the foreign keys record_id and treatment_id which means a treatment can only appear once per record and the quantity column handles repeats.

Hotel Reservation & Management System



1. How does your ERD represent the relationship between reservations and rooms, considering that a reservation can contain multiple rooms and a room can appear in multiple reservations over time?

my ERD resolves it with a junction entity called reservation_room. reservation_room holds two foreign keys, reservation_id (referencing Reservation) and room_number (referencing Room). Together they form its composite primary key, so the same room can't be added twice to the same reservation. The junction table also stores price_per_night, because the price is agreed for each specific room in each specific reservation, and it belongs to neither entity alone.

2. How does your ERD represent the relationship between reservations and services, considering that a reservation can request multiple services and the same service can be requested in many reservations?

My ERD resolves it with the junction entity Service_request. Service_request holds reservation_id and service_id as foreign keys, along with the data about each request: quantity, request_date, and request_time. It has its own primary key, service_request_id, rather than a composite key. This is because the same service can be requested more than once during a single stay (for example, laundry on two different days), so the pair (reservation_id, service_id) wouldn't be unique.